

13 Speech Perception

We have seen that articulatory gestures have acoustic consequences such that articulations lead to sounds that can be described in terms of frequency, intensity, and duration. The structure of the auditory system and the kinds of transformations that it imposes on the incoming signal have also been discussed. In addition to these factors, an understanding of the way speech is perceived requires consideration of linguistic factors that may play a role in our interpretation of a spoken message, including lexical, syntactic, semantic, and pragmatic sources of information. The study of speech perception addresses the way in which the listener analyzes the acoustic patterns and how this information is interpreted by the auditory system in order to recognize the message intended by the speaker. In this chapter, we will focus on the kinds of acoustic properties and auditory constraints that are thought to be important in the perception of individual speech sounds. It is fair to say that most of the research in speech perception has dealt with these issues at the level of individual segments or syllables. Only recently have researchers begun to investigate the perception of larger units such as words and phrases, and, consequently, to consider the role of meaning and syntactic class.

The process of perceiving speech could be thought of as the conversion of a continuous acoustic signal into a set of discrete units. Over the years, many different candidates have been proposed for these units, including features (phonetic properties which languages combine in different ways to construct their phoneme inventories), segments, syllables, and words. There is, as yet, no consensus as to what the basic unit of speech perception is, and it may well be the case that all these candidates play a crucial role at different levels of analysis. Regardless of the exact nature of these linguistic units, they are typically thought to follow each other in time. However, this *linearity* is not often observed in the acoustic signal. The linearity problem relates to the fact that the speech signal can rarely be segmented in a way that corresponds to perceived individual phonetic units. While the process of speech production at some level presumably involves a pre-planned sequence of discrete units or targets, the physiological realization of this sequence results in nearly continuous motion of the articulators. Consequently, the acoustic events that result in the perception of an ordered string of phonemes, for example, do not always occur in a sequential linear order.

While the lack of linearity provides a challenge to theories of speech perception, consensus is emerging about the nature of the acoustic properties that contribute to the perception of individual speech sounds or classes of speech sounds. We will start by discussing perceptual cues to vowels. Next, our overview of perceptual cues to consonants proceeds in terms of manner of articulation. It must be kept in mind that the literature on speech perception is vast, and a comprehensive overview could fill a complete handbook. We will focus here on major and well-established findings.

13.1 Vowels

Vowels are among the perceptually most salient speech sounds. They are typically voiced, relatively loud since they are produced with a comparatively small degree of constriction, and relatively long in duration. The most important acoustic cue to vowel quality is the location of the formant frequencies. As discussed in Section 9.5.1, each vowel is produced with a distinct position of the articulators that results in a distinct location of the formant frequencies. At first glance, vowel perception may therefore seem straightforward: In order to perceive a vowel, all the listener needs to do is extract the location of the first two or three formant frequencies and map this pattern onto an internalized vowel category. For example, based on the information that the location of the first two formant frequencies is 300 and 2,300 Hz, respectively, a native speaker of English may perceive that pattern as the vowel /i/. Likewise, formant frequency values of 800 and 1,100 Hz may be interpreted as /a/.

However, phonetic context, speaking rate, and speaker size can all affect a vowel's specific formant frequency pattern. Variations in phonetic context – that is, in the actual segments that flank the vowel – and speaking rate both contribute to **vowel undershoot**. Undershoot describes a situation in which the formant frequencies do not reach their “canonical” values; that is, those values as measured in vowels produced in isolation where the articulators are in their “optimal” positions. For example, in a CVC syllable, the articulators may not reach their canonical position for the vowel due to the articulatory configuration that needs to be attained for the initial and final consonants and/or the speed with which this needs to be accomplished. That is, the articulators “undershoot” their target positions and, hence, the formant frequencies “undershoot” their canonical values. This does not mean that the formant frequencies are all too low – for example, the “undershot” F1 frequency for a high vowel (which has a low F1) is higher. That is, since the F1 target frequency is low, undershoot will result in a higher frequency value.

Vowel perception also becomes more complicated when considering vowels spoken by a variety of speakers. Since speakers vary in terms of vocal tract length

and since vocal tract length is the primary determinant of formant frequency location, a given vowel will not have the same formant frequencies across a variety of speakers. For example, while a male speaker may produce the vowel /i/ with an F1 of 300 Hz and an F2 of 2300 Hz, a female speaker's /i/ may have an F1 of 400 Hz and an F2 of 2800 Hz. But even with a group of male speakers from the same dialectal area considerable variety exists in the location of formant frequencies. Figure 13.1 shows F1 and F2 values for the English monophthongs as produced by 45 males (Hillenbrand et al. 1995). If each vowel category were uniquely represented by the location of the first two formants, there should be no overlap among them. However, certain vowel categories can clearly be seen to show overlap.

The listener must assign these different acoustic patterns to the same phonemic category, a process that psychologists refer to as **equivalence classification**. The

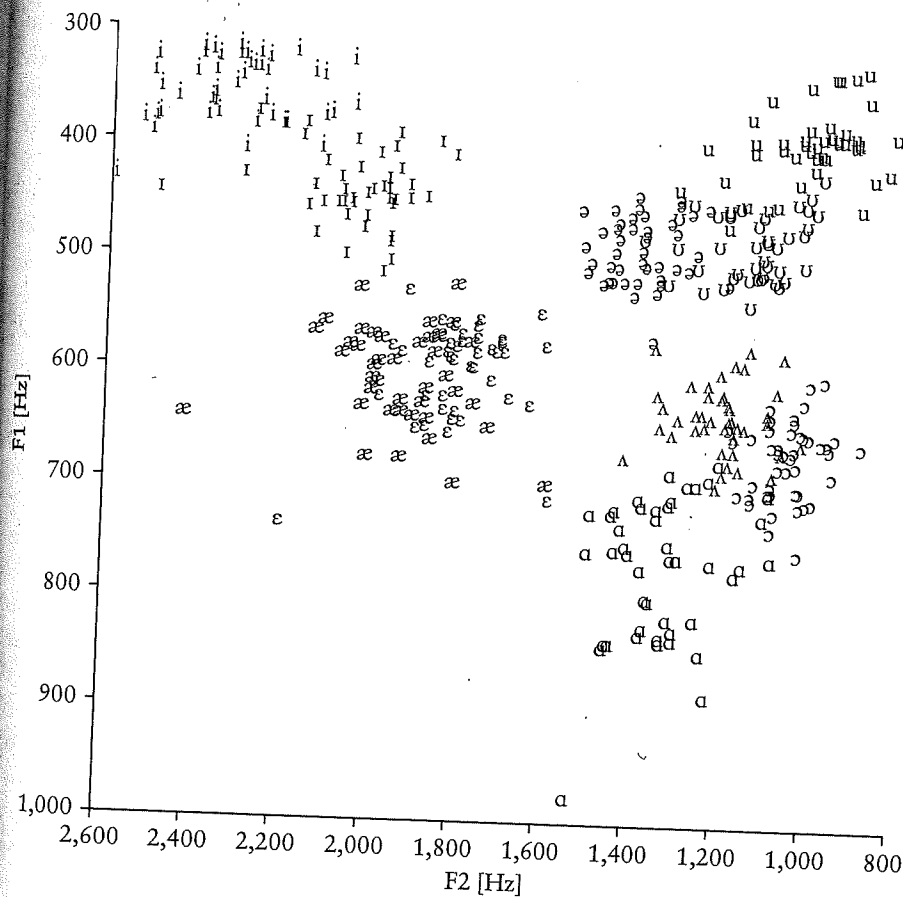


Figure 13.1 Vowel space of Hillenbrand et al. (1995), based on speech by 45 adult males.

process by which the listener is hypothesized to compensate for these variations due to vocal tract size is known as vocal tract **normalization**. The basic idea is that the listener uses certain attributes of the acoustic signal to estimate the length of a speaker's vocal tract in order to determine the frequency parameters of that speaker's vowel space. Normalization is a procedure for factoring out systematic covariation in acoustic properties so that the apparent overlap among vowel categories can be reduced or eliminated.

13.1.1 *Extrinsic versus intrinsic normalization*

While there are many models of vocal tract normalization, two main approaches stand out: **extrinsic normalization** and **intrinsic normalization**. According to a purely *extrinsic* model of normalization, a frame of reference is established from information distributed across the vowels of a single speaker. In other words, surrounding context is thought to allow the listener to calibrate a vowel space for a given speaker such that subsequent vowel productions can be perceived with reference to that space. Evidence for this notion was first presented in a classic study by Ladefoged and Broadbent (1957). These researchers created six synthetic versions of the sentence *Please say what this word is*, each with a different range of F1 and F2. They also created four versions of a target word "bVt" (i.e. a /b/ followed by a vowel and a /t/, e.g. *bit*, *bet*, *bat*, etc.). When presented with these test words in isolation, listeners' percepts shifted from *bit* to *bet* to *bat* to *but* as F1 increased in the vowel. However, when these same words were appended to the precursor sentence, perception of a given target word changed as a function of the formant frequency range in the precursor sentence. For example, the target *bit*, perceived as *bit* in isolation, would be perceived as *bet* when preceded by the precursor in which F1 was relatively low. There was thus a contrast effect, whereby a low F1 in the precursor would make listeners perceive a given F1 in the target as relatively higher (i.e. a lower vowel), thus changing the percept from /t/ to /e/.

While these results provide compelling evidence for some form of extrinsic normalization, they raise questions about the kind and extent of precursor information that is most effective in the calibration of a speaker's vowel space. Perhaps the strongest evidence against the notion of extrinsic normalization is the fact that vowels are perceived with very high accuracy even when different speakers' voices are randomly mixed. In fact, vowel perception under these mixed speaker conditions is as accurate as when each target vowel is preceded by instances of the three **point vowels** /i, u, ɑ/ produced by the same speaker (Verbrugge et al. 1976). The point vowels have been claimed to provide particularly useful information for the process of calibration of the vowel space for a number of reasons: they represent extreme positions in articulatory space, they represent extreme frequencies in acoustic space, their formant frequencies are not substantially affected

by moderate changes in articulatory position, and they are the only vowels for which a formant pattern can be related to a unique vocal tract area function. However, Verbrugge et al. (1976) showed that presentation of these vowels does not seem to improve vowel perception. That is, listeners' identification of vowels is highly accurate, regardless of whether or not they were preceded by instances of the point vowels.

Any theory of extrinsic normalization also begs the question of how successful perception can begin if the perception of a vowel is dependent on that of a previous vowel. One possible answer is that it would take only one vowel to be correctly perceived. The vowel /i/ is the most likely candidate, since Nearey (1978) pointed out that the [i] of any speaker can always be identified as an instance of the vowel /i/ because of its unique formant pattern. In other words, a single token of the vowel /i/ by a given speaker would immediately allow for the estimation of that speaker's vocal tract length. This is an example of an approach known as **point normalization** since it requires only one known point in a speaker's vowel space. A variant is **range normalization**, which requires at least two known vowels.

An alternative to extrinsic normalization is *intrinsic* normalization. According to this view, all the information necessary to identify a vowel is contained within the vowel itself. Overlap among vowel categories is thought not to exist, or at least to be significantly reduced, when the acoustic properties of vowels are appropriately evaluated. This approach typically seeks to derive vowel representations that are similar to those that the human auditory system derives. As discussed earlier, the auditory system imposes a number of non-linear transformations on the speech signal, especially in the frequency and amplitude domains (see Section 12.4.3). Traditionally, vowels have been represented in an acoustic space with a linear frequency scale. More recently, researchers have begun to use more auditorily based scales for frequency (e.g. Bark or ERB scales) as well as loudness (e.g. phon or sone scales). The use of these kinds of scales does seem to substantially reduce the overlap that is traditionally observed among vowel categories (e.g. Syrdal and Gopal 1986). But it should be kept in mind that simply transforming a linear scale to, for example, a logarithmic scale does not change the order of points along a scale: what is "lower" on a linear scale is also "lower" on a logarithmic scale; only the distances between points change. And when the average distance between points (which might represent vowel frequencies) becomes larger after transforming to another scale, it does not necessarily mean that the vowels can be better separated, since only the way the distances are measured has been changed and not the position of the vowels relative to each other.

In addition to the use of more auditorily based scales, researchers have also explored different ways of combining information that is provided by the location of individual formant frequencies. For example, Miller (1989) added F_0 and F_3 to the traditional parameters of F_1 and F_2 to represent vowels in a three-dimensional space whose axes were defined as follows: $x = \log (F_1/F_0)$, $y = \log (F_2/F_1)$, and

$z = \log (F3/F2)$. This approach grew out of formant ratio theory according to which vowel quality depends on the ratios between formant frequencies rather than their absolute values. The idea is that a change in vocal tract length changes all formant frequencies for a given vowel by the same proportion. Thus, computing ratios can factor out differences in vocal tract length. This approach also serves to separate vowel categories quite well.

We started this section by describing normalization as a process that removes idiosyncratic information from the speech signal to obtain abstract prototypical, non-overlapping, categories. Recent research, however, suggests that categories that retain such information may offer certain advantages. In **exemplar-based** theories, a perceptual category (e.g. a consonant or vowel) is thought to consist of all experienced instances of the category. An exemplar consists of an association between a set of acoustic or auditory properties and a set of category labels. Since all relevant acoustic (e.g. detailed spectra) and linguistic and indexical information (e.g. linguistic category, speaker gender) is retained in the exemplar, new items can be categorized by reference to any of these dimensions. The assumption is that "clouds" of exemplars that comprise individual speech categories tend to occupy more or less unique regions of the higher-dimensional space that is generated by all of this information. Johnson's (1997) results from simulations with an exemplar-based model of vowel identification show highly accurate vowel recognition without any explicit form of speaker normalization. The extent to which exemplar-based models can accommodate other sources of variability in the speech signal is still under debate (see Johnson 2003).

Finally, it is important to note that the approaches to vowel representation mentioned thus far are all based on formant frequencies measured in the middle of the vowel. This midpoint measurement is as far away as possible from flanking segments and is considered to be least affected by contextual effects such as coarticulation. In other words, the formant frequencies measured at vowel midpoint may provide the "purest" representation of a vowel. However, when focusing on the steady-state portion of the vowel, other potentially important information is lost. Specifically, dynamic properties such as the duration of the vowel and the formant transitions into or out of the vowel may provide important cues to vowel quality. Indeed, some experiments (e.g. Strange et al. 1976; but see Macchi 1980) have shown that vowels in a CVC context were more accurately perceived than isolated vowels, suggesting that listeners do not exclusively base their vowel perception on steady-state information but use dynamic information as well. Further support for the importance of dynamic information comes from a perception experiment by Hillenbrand and Nearey (1999). Listeners were presented with synthetic "hVd" (e.g. *heed*, *hid*, *head*) stimuli for identification. Two synthetic versions were created: one was a close copy of natural productions; the other contained no formant transitions since the formant values were set to those measured at the "steadyest state" of the vowel. Results showed that vowel identification was significantly worse for the stimuli without transitions (74 percent) than for the stimuli in which spectral change was preserved (89 percent).

13.2 Consonants

Perception of consonants is often considered more complex than vowel perception since the acoustic characteristics of consonants are thought to be more variable. Among the sources of variability that may influence the acoustic characteristics of consonants are vocal tract size, phonetic context, speaking rate, and stress. We will begin with approximants since they are most similar to vowels.

13.2.1 Approximants

Approximants are produced with a small constriction. The tongue approximates the passive articulator but does not come very close. As a result, the acoustic properties of approximants resemble those of vowels in many respects. Perceptually, the formant transitions into and out of the approximant are an important cue to place of articulation (see Section 10.2.1). The onset frequency of the F2 transition has been shown to be a primary cue to the distinction between /w/ and /j/. Since /w/ has a bilabial constriction (in addition to its velar constriction), all formant frequencies are a bit lower (see Section 10.2.1) and a low F1 and F2 are its main perceptual cues. Since /j/ has a palatal constriction, the front cavity is shorter and a high F2 is its primary cue. The distinction between /ɹ/ and /l/ seems to be carried by a number of cues. The most obvious difference between these two sounds is the location of F3: /ɹ/ has a very low F3, well below that for /l/. However, the onset frequency of F2 (low for /ɹ/, higher for /l/) and the transition duration of F1 (long for /ɹ/, shorter for /l/) have also been shown to contribute to the perception of the /ɹ/-/l/ distinction.

Research on the cues to approximant manner of articulation has focused on the distinction between the plosive /b/ and the approximant /w/. One important cue to this distinction is the duration of the formant transitions, particularly F2. /b/ has short transitions while /w/ has relatively long transitions. Using synthetic speech, a reasonable-sounding /b/ can be changed into an acceptable /w/ by lengthening the F2 transition duration while keeping all other parameters constant. Any time a distinction is carried primarily or exclusively by a temporal parameter, variations in speaking rate are likely to affect the perception of this distinction, as indicated in a change of the **category boundary**, that is, that value of one or more acoustic parameters (in this case, F2 transition duration) that listeners identify 50 percent of the time as a member of one category (e.g. /b/), and 50 percent of the time as a member of the other (e.g. /w/). (See Section 13.3.1 for a more detailed discussion of category boundaries.) Indeed the category boundary between /b/ and /w/ has been shown to shift as a function of speaking rate: an increase in speaking rate, experimentally induced by shortening the vowel following the initial consonant, results in an increase in the number of /w/ responses (Miller

and Liberman 1979). This is thought to occur because a shorter vowel makes a given transition duration seem relatively longer. This is an example of the context-dependent nature of acoustic cues.

However, manipulating F2 transition duration as the sole cue to the /b/-/w/ distinction may be a simplification. As discussed in Section 10.4.1, the difference between stops and glides in terms of rapid versus gradual release affects amplitude as well as transition duration. For example, /b/ has a sudden increase in amplitude at the moment of release while /w/ shows a very gradual change. In natural speech, both transition duration and amplitude, as well as a variety of more subtle cues, all contribute to the /b/-/w/ distinction. And when this distinction is signaled by all these cues simultaneously, the effect of speaking rate has been shown to substantially diminish or disappear altogether (Shinn et al. 1985), although the rate effects tend to come back in the presence of background noise (Miller and Wayland 1993).

13.2.2 Fricatives

Fricatives are relatively long and are quite steady throughout their duration. Two defining features of fricatives are the fricative noise itself and the transitions from the fricative into the following vowel. Acoustically, spectral peak location and relative amplitude (see Section 10.2.2) serve to distinguish fricatives in terms of place of articulation.

Perceptually, the role of spectral peak location has been investigated in several studies. Most notably, Heinz and Stevens (1961) synthesized fricatives by systematically varying the location of the spectral peak from 2 to 8 kHz. Listeners' responses showed that fricatives with a peak below 3 kHz were identified as /f/; fricatives with a peak between 4.5 and 6.5 kHz were identified as /s/, and those with a peak above 6.5 kHz were perceived as /f/ or /θ/. However, listeners were unable to distinguish /f/ from /θ/. Research investigating the role of vocalic context on the perception of /s/ and /f/ has also shown that an /s-/f/ continuum can be successfully created by varying spectral peak location (e.g. Mann and Repp 1980; Whalen 1981). In sum, these studies indicate that spectral peak location is a reliable perceptual cue to the distinction between sibilant /s, ʃ/ and non-sibilant /f, θ/ fricatives, and peak location also serves to perceptually distinguish /s/ from /f/.

Recent research suggests that relative amplitude is also a perceptual cue to place of articulation in fricatives. In particular, Hedrick and Ohde (1993) showed that the amplitude of friction relative to the amplitude of vowel onset affects listeners' perception of /θ/, /s/, and /f/. It is as yet unknown whether relative amplitude is a cue in the perception of distinctions *within* the class of non-sibilant fricatives.

Evidence in support of the perceptual significance of other properties of the friction noise is mixed at best. Behrens and Blumstein (1988) showed that noise amplitude alone is not a reliable cue to fricative perception. In addition, noise duration does not seem to be an important perceptual cue to fricative place of articulation.

Finally, support for the role of transition information is provided by Harris (1958) and, more recently, Nitttrouer (2002). Harris recorded fricative-vowel syllables consisting of all English fricatives and four vowels. She then created stimuli for a perception experiment by cross-splicing¹ the friction noise from one syllable onto the vocalic portion of another syllable. Results showed that the sibilants were identified on the basis of the friction noise: combining the noise portion of /s, z, ʃ, ʒ/ with any of the vocalic portions resulted in perception of /s, z, ʃ, ʒ/ respectively. However, perception of /f, θ/ was largely determined by the vocalic portion. For example, stimuli with /f/ or /θ/ noise were perceived as /f/ when followed by vocalic portions excised from /f/, and perceived as /θ/ when followed by any other vocalic portion. Based on these results, Harris (1958) assumed that listeners first decide, on the basis of the friction noise, whether they heard /s, ʃ/ or /f, θ/, and that they then use the vocalic portion to decide whether they heard /f/ or /θ/.

This notion is further supported by the finding that listeners' perception of /f, θ/ was significantly above chance when presented with only the fricative-to-vowel transitions, while perception of /s, ʃ/ was not. In addition, perception of /s, ʃ/ did not suffer from removal of the transitions. In sum, there is evidence that formant transitions play a role in the perception of the non-sibilant fricatives.

The voicing distinction is primarily cued by the presence versus absence of vocal fold vibration during the fricative, seen acoustically as the presence versus absence, respectively, of low-frequency energy. While voiceless fricatives are generally longer than voiced fricatives, duration per se does not seem to be a sufficient cue to fricative voicing. Finally, fricatives are distinguished from other obstruents in terms of their onset characteristics: fricatives have a long rise-time, plosives have a short rise time, and affricates fall in between. As long as the onset characteristics are preserved, even fricatives that have been severely shortened are still perceived as fricatives (Jongman 1989).

13.2.3 Nasals

Nasal stop consonants have a number of defining characteristics, including the pattern of formants and anti-formants during the nasal murmur, and the formant transitions into and out of the nasal. The cues to the nasal *manner* of articulation are relatively straightforward while those to place of articulation are more complex. The nasal murmur portion contains several cues to the nasal manner of articulation, including weak formants with large bandwidths, a low-frequency resonance (at approximately 300 Hz) known as the nasal formant, and anti-formants (see Section 10.2.4). The presence of this murmur is a sufficient cue for the perception of the nasal manner of articulation. When an oral vowel is followed by a nasal, the velum is typically lowered during the latter portion of the vowel. The weakening of the higher formants in the vowel spectrum as the result of the opening of the velar port also provides a perceptual cue to the upcoming nasal.

The perceptual cues to *place* of articulation reside both in the nasal murmur and the formant transitions. While early research (e.g. Liberman et al. 1954; Maleco 1956) concluded that the transitions, but not the murmur, carried cues to place more recent studies indicate that listeners use both sources of information. Specifically, while identification of place of articulation seems comparable on the basis of either the nasal murmur or formant transitions, identification is superior when listeners are presented with a few pulses of both murmur *and* transitions (Kurowski and Blumstein 1984; Repp and Svastikula 1988).

13.2.4 Plosives

Acoustically, plosives are characterized by a number of events: consonant closure, release burst, and formant transitions (both from a preceding segment into the plosive and from the plosive into a following segment). Each of these properties can be analyzed in terms of its frequency, amplitude, and duration. Voice onset time (VOT) is an additional temporal parameter that serves to distinguish plosives. In terms of manner of articulation, plosives are unique because of their closure portion, release burst, and rapid formant transitions. Research has shown that listeners correctly identify them as plosives when presented with only the first 10–20 ms of CV syllables. Plosive voicing can be cued by a number of parameters. Relative to voiced plosives, voiceless plosives typically have a longer closure portion, typically with no or only brief glottal pulsing, a more intense release burst, a positive VOT, and higher F_0 and F1 onset frequencies in the following segment.

Most research on the perception of plosives has focused on cues to place of articulation. Two acoustic characteristics containing information about place are the frequency of the release burst and the formant transition pattern. Over the past 50 years, a great many researchers have attempted to determine if and to what extent these properties, either alone or in conjunction, enable listeners to identify a plosive's place of articulation. Early research indicated that burst frequency per se is not a very reliable cue to place of articulation. Using synthetic speech, so that the artificially generated speech allowed researchers to vary one acoustic property while keeping all others constant, Liberman et al. (1952) created a series of release bursts, ranging in frequency from 360 Hz to 4,320 Hz. These bursts were followed by a series of two-formant (F1 and F2) patterns representing seven different English vowels. Listeners were presented with these synthetic stimuli and asked to indicate whether they thought the initial consonant was /p/, /t/, or /k/. The results are shown in Figure 13.2. While the bursts consistently elicited plosive responses, place of articulation identification of the bursts was heavily influenced by the vowel context. In general, high-frequency bursts were always identified as /t/, and very low-frequency bursts were predominantly perceived as /p/. This finding matches our discussion of acoustic cues to place of articulation in Section 10.2.3. However, a mid-frequency burst, around 1,400–2,000 Hz, was perceived as /k/ when followed by /e, a, ɔ/ but as /p/ when

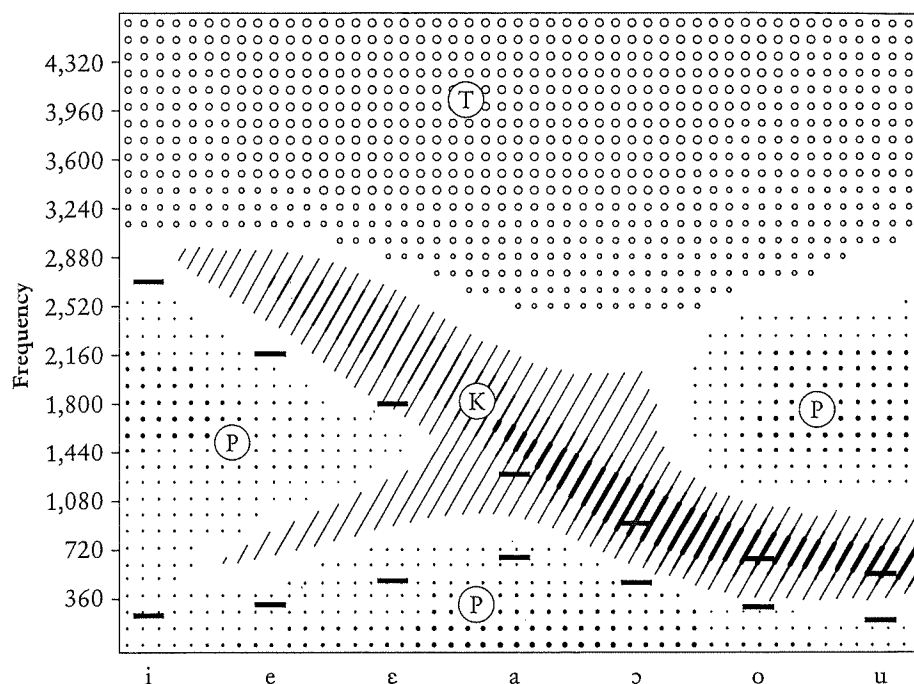


Figure 13.2 Listeners' identification of synthetic stimuli used in a burst perception study, from Liberman et al. (1952). Frequency (in Hz) is represented along the vertical axis. Open circles indicate the burst frequencies for which a /t/ was perceived, closed circles indicate a /p/, and diagonal lines a /k/. Thickness indicates the strength of the perception. Horizontal lines mark the location of the formant frequencies (F1 and F2) of the appropriate vowels.

Used with permission of Alvin M. Liberman, Pierre C. Delattre, Franklin S. Cooper, and the University of Illinois Press, from *The American Journal of Psychology*, 65, 497 (1952). Copyright, 1952 by the Board of Trustees of the University of Illinois.

followed by /i, e, o, u/. Burst frequency did not seem to be a reliable cue to place of articulation, particularly for velars.

Subsequent experiments used cross-spliced natural speech stimuli to examine the perception of place of articulation. For example, Cole and Scott (1974) recorded the English voiced and voiceless plosives before the vowels /i, a, u/. They then removed the consonant from one vowel context and cross-spliced it onto a different vowel context. For example, the burst and aspiration of /ti/ "t(i)" was spliced onto the vocalic portion taken from /tu/ "(t)u", and vice versa. Cole and Scott (1974) showed that identification of bilabials and alveolars was highly

accurate regardless of changes in vowel context. Perception of velars, however, was much poorer. Applying the same technique to labial, alveolar, and velar voiced plosives spoken before nine different vowels, Dorman et al. (1977) reported poor identification of place of articulation.

Some studies presented for identification only the release bursts of naturally produced plosive-vowel syllables and reported that listeners could identify place of articulation with reasonable to good accuracy (e.g. Winitz et al. 1971; Kewley-Port et al. 1983). Indeed, Tekieli and Cullinan (1979) showed that listeners' identification of place of articulation was significantly better than chance when presented with only the first 10 ms of voiceless plosives followed by eight different vowels. Overall, this body of research suggests that burst frequency may be a necessary but not a sufficient cue to the perception of place of articulation in stop consonants.

The role of formant transitions as perceptual cues to place of articulation has also been investigated in a variety of experiments. In a complementary study to the one on the role of burst frequency described above, Liberman et al. (1952) created synthetic two-formant patterns (without bursts) in which the F2 transition was systematically varied. Perceptual results indicated that most listeners perceived a rising F2 as a bilabial plosive but that a falling F2 was heard either as an alveolar or velar plosive, depending on the vowel.

Another method of evaluating the perceptual role of formant transitions is to remove the burst from naturally produced plosive-vowel syllables, leaving only the formant transitions. Results from such studies indicate that identification rates of place range from moderate to good for voiced plosives but are substantially worse for voiceless plosives. (e.g. LaRiviere et al. 1975; Dorman et al. 1977).

The most careful and comprehensive study to date examining the contribution of bursts and formant transitions to plosive identification was conducted by Smits et al. (1996) for the Dutch plosives /p, t, k, b, d/ followed by the vowels /i, y, a, u/. Using digital cross-splicing and filtering techniques, burst-only, burstless, and cross-spliced stimuli were created. Average recognition rates for the burst-only and burstless stimuli are shown in Table 13.1.

Cross-spliced stimuli were created by combining the burst of a plosive at a given place of articulation (e.g. the burst from /pa/ "(p)a") with everything but the burst of a plosive at a different place of articulation (e.g. from "(t)a" or "(k)a"). The resulting stimuli are known as **conflicting-cue** signals because they combine cues to conflicting percepts. In this case, the burst signaled one place of articulation while the formant transitions signaled a different place of articulation. Responses to the voiceless cross-spliced stimuli corresponded to the burst identity in 49 percent of the cases and to the transitions in 43 percent of the cases. In addition, the /k/-burst was a stronger cue to place than the /p/- and /t/-bursts and /p/, /t/, and /k/ transitions were equally strong cues to place of articulation. For the voiced cross-spliced stimuli, responses corresponded to the burst identity in only 26 percent of the cases and to the transitions in 74 percent of the cases. The /d/-burst was a stronger cue than the /b/-burst.

Table 13.1 Correct identification (in percentages) of place of articulation, averaged across vowel contexts, for burst-only and burstless stimuli (after Smits et al., 1996)

<i>Percentage correct identification of . . .</i>							
Stimulus	[p]	[t]	[k]	Mean	[b]	[d]	Mean
Burst-only	80	50	91	74	65	57	61
Burstless	95	64	47	69	96	87	92

In sum, this study confirms and extends our understanding of the relative importance of bursts and transitions as cues to place of articulation in plosives. While cue strength varies with place of articulation, vowel context, as well as speaker, a number of generalizations emerge. In general, the burst is the strongest cue to place perception in front vowel contexts while the transitions are most important in back vowel contexts. Moreover, the bursts of voiceless plosives provide a stronger cue than those of voiced plosives while the transitions of voiced plosives provide more information than those of voiceless plosives.

As mentioned in Section 10.4.1, the fact that individual acoustic properties such as burst frequency and formant transition frequency often show a high degree of context-dependency has led some researchers to consider the role of global, rather than local, properties as perceptual cues to place of articulation. For example, we mentioned in Section 10.4.1 that Lahiri et al. (1984) found that the change in distribution of high-frequency energy as compared to low-frequency energy between consonant onset and the onset of the following vowel provides a stable cue. Perception experiments in the same study confirmed that listeners integrate these two cues, burst onset spectrum and vowel onset spectrum, to make a phonetic decision.

13.3 Contributions of the Motor Theory of Speech Perception

As discussed in Section 10.4, Liberman and colleagues conducted a great many careful experiments in their search for correlates of phonemic distinctions. However, they failed to find acoustic correlates that were stable in the face of contextual variability. The lack of acoustic invariance raised the question of which other sources of information listeners use to perceive speech. Liberman and colleagues concluded that it must be articulatory information and formulated

the **motor theory of speech perception**. According to *motor theory*, as it is commonly abbreviated, speech perception is more closely related to articulation than to acoustics. Motor theory claims that speech is perceived by reference to articulation: the articulatory movements and their sensory effects mediate between the acoustic stimulus and its perception. Motor theory essentially hypothesizes that listeners interpret the received auditory patterns in terms of articulatory patterns that would produce those auditory patterns. Thus, the invariant cues to speech sounds are the articulations that produce them. One of the attractive features of this theory was its direct link between the perception and production of speech. The assumption that the processes by which speech is produced and perceived have much in common may be intuitively more appealing than a view under which the two are unrelated. Early articulatory studies provided support for the theory that articulatory positions and movements classify a speech sound despite its acoustical variation (Liberman 1957). Unfortunately, it was also observed that the same speech sound can be realized by quite different articulatory movements (MacNeilage 1970), hence the sequence of articulatory movements was as variable as the speech signal itself. The motor theory was therefore modified, such that the motor commands to the articulators, rather than the movements themselves, classify speech sounds (Liberman 1970). Consequently, the listener's knowledge about the effect of the motor commands on the vocal tract serves as units of the perception process (Liberman 1982). The theory was further extended to the **revised motor theory**, namely that the **intended gestures** are perceived and thus, without even being realized, enable the perception of speech (Liberman and Mattingly 1985). Listening, in this view, is the perception of the intended gestures that lead to the coarticulated and highly variable speech signal.

Another gestural account of speech perception is **direct realism** (Fowler 1986, 2003). Direct realism is a theory of perception in general, not just auditory perception. In terms of speech perception, motor theory and direct realism both posit that the immediate objects of perception are articulatory gestures. However, while for motor theory the objects of perception are intended gestures, or the commands that would give rise to these gestures, direct realism assumes that these objects are actual vocal tract movements. Analogous to the way in which our eyes perceive objects rather than the physical reflections of light, our ears perceive articulatory gestures instead of the physical changes in frequency, amplitude, and duration of the speech signal. Such a view requires that vocal tract movements are uniquely recoverable from the acoustic signal, which does not seem to be the case since there are multiple articulatory ways to produce a given acoustic signal.

The major criticism of the original motor theory was the missing explanation of how the articulatory positions and movements can be perceived from the speech signal and how a child learns this conversion. Even in the revised motor theory, no actual mechanism is proposed. Recent neurological research may support the possibility of motor recruitment in speech perception posited by motor theory. Rizzolatti and colleagues discovered a group of neurons – **mirror neurons** – in monkeys that are active in the matching of observed events to similar internally

generated actions. Specifically, these mirror neurons discharge both when monkeys perform an action such as tearing a piece of paper and when they only hear the sound of paper tearing (e.g. Rizzolatti and Arbib 1998; Kohler et al. 2002). This is an exciting finding that will stimulate further research on the link between the production and perception of speech. It is currently too early to tell whether the notion of mirror neurons vindicates motor theory. For example, neurons responding to the sound of paper being torn could be shaped by associative learning as part of a system that evolved for rapid responses to environmental situations. For now, it is clear that motor theory has inspired an impressive body of research, by both its proponents and critics, that has led to important insights into the nature of speech perception.

13.3.1 Categorical perception

Perhaps the best-known finding by Liberman and colleagues is that of **categorical perception**. Categorical perception refers to the fact that stimuli, equally spaced along some physical continuum, are perceived as belonging to one or another perceptual category instead of as varying as a function of their physical values. In the case of speech, the phenomenon of categorical perception refers to the way in which certain distinctions are perceived. Categorical perception requires tests of listeners' abilities to identify and to discriminate speech sounds. In general, people are much better at **discriminating** sounds (i.e. hearing the difference between a pair of sounds) than at **identifying** (labeling; i.e. saying "this is a *C sharp* note" and "that is a *B*") them. For example, listeners can discriminate hundreds of different pitches yet identify fewer than 10.

An example will make this clear. Consider the voicing distinction in English syllable-initial plosives. Voiced plosives typically have a short positive VOT while voiceless plosives have a long positive VOT (see Section 10.2.3). Using either synthetic or manipulated natural speech, it is possible to create a series of stimuli that systematically differ in terms of their VOT. The first member could have a VOT of 0 ms, the next member a VOT of 10 ms, and so on until the last member with a VOT of 80 ms. This would result in a voicing continuum, say from /da/ to /ta/, with nine members. Several repetitions (for example 10) of each member are then created and the order of these 90 tokens is **randomized**. Randomization simply arranges the stimuli in arbitrary order. The randomized stimuli are then presented to listeners who have to identify them as either /da/ or /ta/. In other words, a listener may be presented with the 40 ms VOT stimulus on the first trial, the 0 ms VOT stimulus on the second trial, the 60 ms VOT stimulus on the third, and so on. We would expect the 0 ms and 80 ms VOT stimuli to be identified as /da/ and /ta/, respectively. How would the other members be identified? A reasonable guess would be that as VOT increases, the number of /da/ responses decreases in a linear fashion, as illustrated in Figure 13.3. However, typical results of such an identification experiment look more like Figure 13.4.

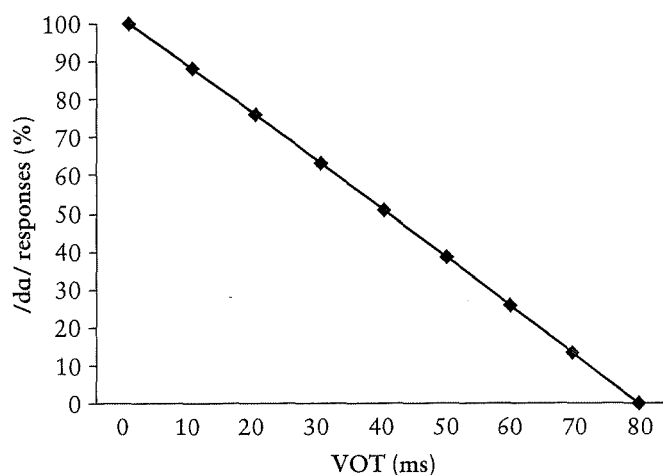


Figure 13.3 Expected result of identification experiment in which VOT is systematically varied.

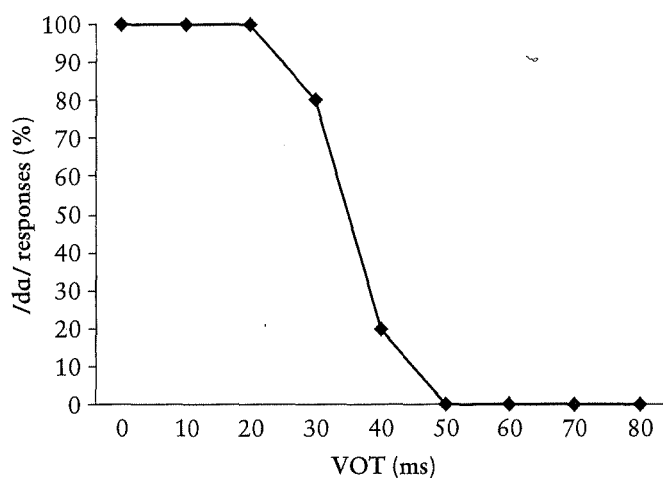


Figure 13.4 Obtained result of identification experiment in which VOT is systematically varied in 10-ms steps from /da/ to /ta/.

The change from /da/ to /ta/ is not gradual but abrupt or categorical. The listener divides the stimuli into two sharply defined categories, "da" and "ta." The category (or phoneme) boundary between these two categories, computed as the VOT value at which /da/ (and /ta/) identification is at 50 percent, corresponds to a VOT of approximately 35 ms. Notice that there is very little difference in the identification of stimuli that are assigned to the same category. For example, both

the 0 ms and 10 ms VOT are consistently identified as /da/. However, the identification of two stimuli on opposite sides of the boundary is very different, even though they are separated by the same physical distance. That is, the 30 ms VOT stimulus is perceived as /da/ 80 percent of the time while the 40 ms VOT receives only 20 percent /da/ identifications.

In addition to a sharp identification function, categorical perception also involves a characteristic discrimination function. Since a prerequisite of categorical perception is that discrimination is no better than identification, a discrimination experiment will have to be conducted as well (otherwise this pattern of results cannot be called categorical perception). There are several ways of conducting a discrimination experiment and we will focus on the most commonly used method, the ABX or odd-ball task. In this particular version of a discrimination experiment, three stimuli, A, B, and X, are presented on each trial. A and B are always different stimuli while X always matches either A or B. The listener's task is to indicate whether X matches A or B. For example, with A and B chosen such that they are always separated by one continuum member (known as "two-step discrimination" because two steps are necessary to go from A to B), there would be seven possible combinations of A and B for the 0–80 ms VOT continuum discussed earlier (0 and 20 ms VOT, 10 and 30, 20 and 40, 30 and 50, 40 and 60, 50 and 70, and 60 and 80). Thus, a trial could consist of the triad 0–20–20 and the listener has to indicate whether the final 20 ms VOT stimulus sounds more like the first 0 or the second 20 ms VOT. Again, 10 or so repetitions of each triad are created and randomly presented to listeners for discrimination. If the two stimuli are easy to distinguish, the listeners' accuracy should be at or near 100 percent. If the two stimuli are difficult to distinguish, listeners will identify a given stimulus X half the time as A and half the time as B, resulting in an accuracy of 50 percent. Results from the discrimination experiment are shown in Figure 13.5.

The discrimination function is flat at both ends and has a peak in the middle. There is indeed one stimulus pair that is apparently very easy for listeners to distinguish, as indicated by the peak in the discrimination function, and it consists of the 20 and 40 ms VOT stimulus pair. In other words, listeners are very good at telling these stimuli apart. Recall that in the identification experiment, these stimuli were identified as members of two different categories, "da" and "ta," respectively. The figure also shows that discrimination accuracy hovered around 50 percent for virtually all other stimulus pairs. This indicates that listeners were able to distinguish the two members of a pair only half of the time. In other words, half of the time they thought the members were the same and half of the time they thought they were different. In essence, then, listeners were simply guessing. For these stimuli, performance is said to be at chance level. That is, with only two response categories ("same" or "different"), guessing would result in an accuracy level of 50 percent. Notice that all stimulus pairs for which discrimination was at chance level consisted of stimuli that in the identification experiment had been identified as members of the *same* category.

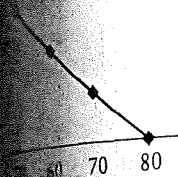


Figure 13.5 Identification function for the /da/ category. The x-axis represents VOT in milliseconds, and the y-axis represents the percentage of /da/ identifications.



Figure 13.5 Discrimination function for the /da/ category. The x-axis represents VOT in milliseconds, and the y-axis represents the percentage of correct discriminations.

...or categorical. The ... "da" and "ta." The ... computed as the ... percent, corresponds to ... very little difference in the ... category. For example, both

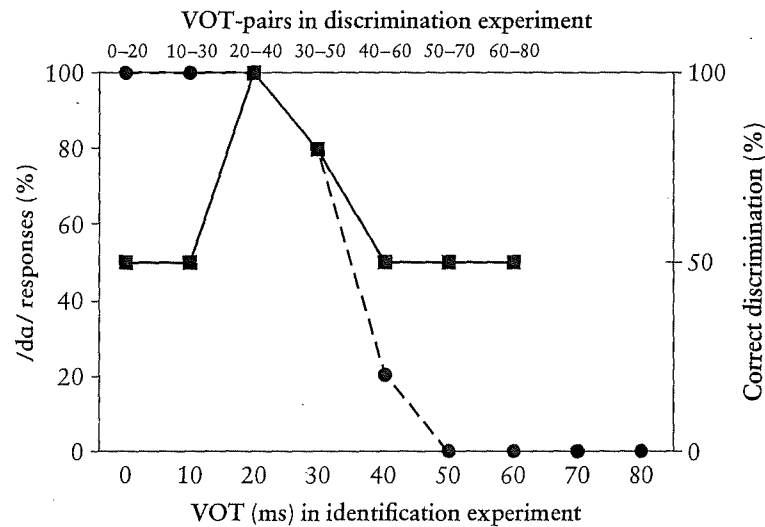


Figure 13.5 Results (solid line) from a two-step discrimination experiment using the same VOT continuum members as in identification. The dashed line again shows the identification results.

In sum, these results show that discrimination is indeed no better than identification, which is rather different from what we said about pitch perception at the beginning of this section. There is no difference in identification of stimuli that are perceived as belonging to the same category. In addition, the ability to discriminate between members of the same category is at chance level (50 percent) while discrimination is very good for the stimulus pair that straddles the phoneme boundary. This pattern of identification and discrimination is the hallmark of categorical perception. In addition to the perception of voicing, perception of place of articulation (e.g. using a /ba-da-ga/ continuum in which the F2 transition is systematically modified) has also been shown to be categorical. Categorical perception of plosives is a very robust finding. It has been extensively replicated across different languages and listeners (for an extensive overview, see Repp 1984).

In contrast, vowel perception seems to be continuous. When a vowel continuum is created, for example from /i/ to /e/ by raising F1 in a number of equal steps, perception of this continuum is not categorical. Instead, it looks more like that in Figure 13.3, displaying a gradual, continuous change. The fact that plosives, but not vowels, are categorically perceived was initially taken as support for motor theory. Perception was seen as mimicking production. Since the plosives /b, d, g/ are produced by categorically different gestures, they are perceived in a categorical way. Vowels, on the other hand, are produced much more continuously, and they are perceived in a continuous way.

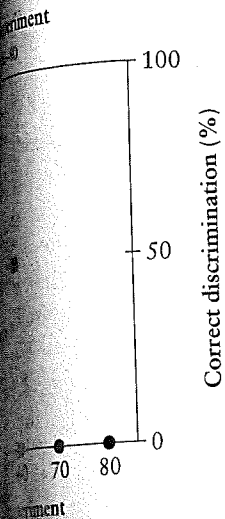
Subsequent research, however, has suggested that this difference in perception between plosives and vowels may not be related to articulatory differences. These two types of sounds also differ in duration: vowels are relatively long and plosives

are relatively short and, maybe more importantly, dynamic. When vowels are shortened to the extent that they are of comparable duration to plosives, their perception becomes categorical as well.

The initial interpretation of categorical perception that listeners can only distinguish stimuli from different categories is probably too strong. The original claim that listeners are not able to distinguish among different stimuli from the same category may have been due to limitations of the experimental task. The use of two additional response measures, **reaction time** and **goodness ratings**, indicates that listeners do perceive differences among members of the same category. *Reaction time* is often used as a measure of amount of processing, particularly in psycholinguistic research. Typically, the more ambiguous a stimulus, the more time it takes for a subject (i.e. a listener in an experiment) to process it, and the time to perform a certain reaction (e.g. pressing a button) will be longer. *Goodness ratings* indicate how good an instance a given token is, as judged by a subject. For example, a subject has to decide whether he hears a /ta/ and has to give a rating from "this is a good /ta/" to "this is a bad /ta/." Alternatively, there could be a line with /ta/ on one end and /da/ on the other and the subject has to make a cross on the line to indicate how close the presented item was to a /ta/ or a /da/; that is, how well a stimulus (the presented item) matched the target (/ta/ or /da/). The initial identification results show that all stimuli within a category are identified as if they are all identical (e.g. 100 percent /da/). However, using reaction time and category goodness ratings, categories have been shown to have internal structure in that stimuli closer to the phoneme boundary are identified more slowly and receive lower goodness ratings. Recent studies using measures such as eye movements and EEG provide additional support for the notion that within-category detail is important in speech perception (e.g. McMurray et al. 2002; Sharma and Dorman 1999). Thus, these kinds of studies have caused us to rethink the original notion of categorical perception. While it is probably efficient to ignore subtle acoustic differences between sounds if they do not signal a difference in phoneme category, that does not mean that listeners are not aware of those differences. In this sense, the phenomenon of categorical perception in which discrimination is not better than identification is a form of equivalence classification (see Section 13.1).

13.3.2 Is speech "special"?

The finding that human adults perceive plosives in a categorical fashion naturally triggered interest in the origin of categorical perception. Specifically, the debate focused on whether categorical perception was an innate attribute, due to the structure of the auditory system, or arose as the result of exposure to language. Researchers recognized that a logical way of addressing this question would be to explore the nature of plosive perception in newborn human infants, since they lacked significant exposure to language. However, it was not until the late 1960s



discrimination experiment
identification. The dashed line

indeed no better than identification about pitch perception at the identification of stimuli that. In addition, the ability to discriminate at chance level (50 percent) that straddles the phoneme boundary is the hallmark of categorical perception. Categorical perception is extensively replicated across experiments (see Repp 1984).

When a vowel continuum is presented in a number of equal steps, it looks more like a continuous change. The fact that categorical perception was initially taken as evidence for categorical production. Since different gestures, they are produced much differently.

The difference in perception of continuous differences. These differences are relatively long and plosives

that a technique was developed that allowed the testing of infants. This technique, known as the **high amplitude sucking (HAS)** paradigm, was first introduced in a classic study by Eimas et al. (1971). Briefly, this paradigm exploits the fact that an infant's sucking rate increases when presented with a novel stimulus. The infant sucks on a pacifier that is wired to a transducer so that the sucking rate can be monitored. A given stimulus is repeatedly presented over loudspeakers. As the novelty of the stimulus wears off, the infant's sucking rate decreases, a process known as habituation. The critical question is what happens when the infant is subsequently presented with a different stimulus: if the infant treats the stimulus as different from the previous one, we would expect the sucking rate to increase; if not, we would expect the rate to remain stable or decrease even further. The infant is therefore engaged in a discrimination task.

Eimas and colleagues tested 4-month-old English-speaking infants to investigate perception of the voicing distinction in English syllable-initial plosive consonants. A synthetic /ba-pa/ continuum was created in which VOT increased from 0 to 80 ms in 20 ms steps. As shown in Figure 13.6, these infants performed very similarly to adults. The leftmost panel indicates that infants were able to discriminate between two stimuli that for adults belong to two distinct categories. Infants clearly differentiate between a 20 ms VOT as one category and a 40 ms VOT as another category. The middle panel shows that infants did not distinguish stimuli that for adults were members of the same category. Thus, infants did not discriminate a stimulus with a 0 ms VOT from one with a 20 ms VOT. Likewise, infants did not discriminate a stimulus with a 60 ms VOT from one with an 80 ms VOT, both of which are categorized by adults as voiceless. Finally, the control condition on the rightmost panel confirms that sucking rate continues to decrease when the presented stimulus does not change.

While these results are sometimes interpreted as showing that infants perceive plosive voicing in a categorical manner, a true demonstration of categorical perception requires the results of both a discrimination and an identification test to converge. While it may prove impossible to obtain identification data from infants, one has to be cautious in the interpretation of the discrimination results. Nevertheless, the infant perception results suggest that the categorical perception observed in adults is not due to linguistic experience. Instead, this behavior seems to be present at birth. Adult-like perception has been shown by infants for a variety of linguistic distinctions in addition to voicing.

The finding that the perception of certain linguistic distinctions did not seem to require linguistic exposure led to the debate about whether or not human speech perception makes use of mechanisms that evolved especially for the processing of speech. This debate, sometimes referred to as the **speech is special** debate, centers around two competing hypotheses, referred to by Kuhl (1989) as the **special mechanism account (SMA)** and the **general mechanism account (GMA)**. According to SMA, speech is perceived directly as the result of specialized, speech-specific mechanisms that exist only in humans. According to GMA, speech perception is based on general auditory and cognitive mechanisms and does not require

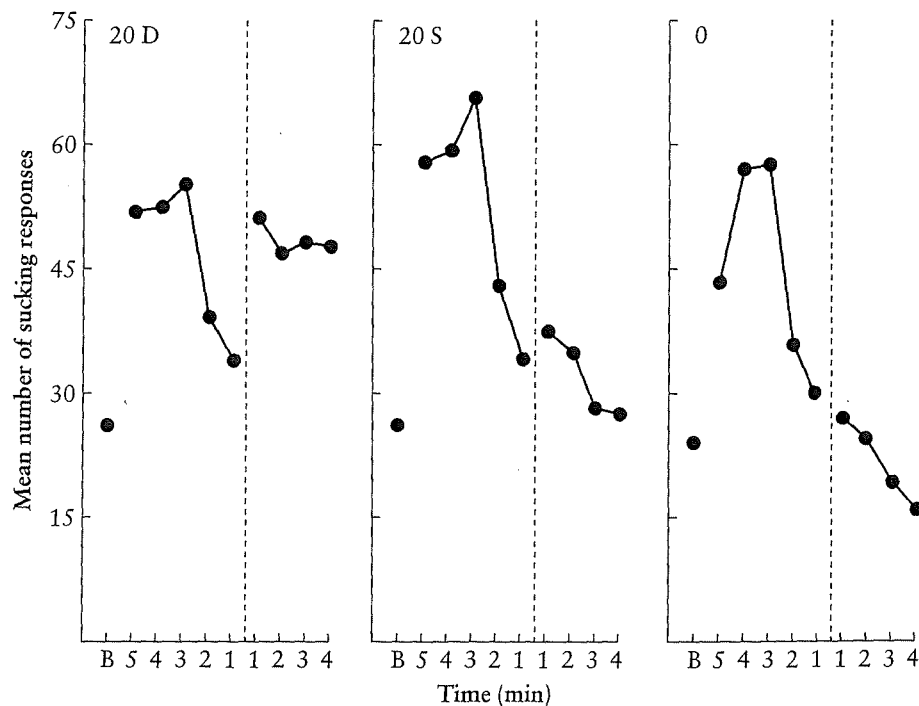


Figure 13.6 Results from Eimas et al. (1971). Mean number of sucking responses for 4-month-old infants as a function of time and experimental condition. The dashed line indicates the occurrence of the stimulus shift, or, in the case of the control group (right panel), the time at which the shift would have occurred. The left panel shows responses to a change from a 20 ms to a 40 ms VOT. The middle panel shows responses to a change from a 0 ms to a 20 ms VOT. The right panel shows responses to no change in VOT. B indicates baseline sucking rate. Time is measured with reference to the moment of stimulus shift and indicates the 5 minutes prior to and the 4 minutes after shift.

Reproduced by permission of the American Association for the Advancement of Science.

innate speech-specific processes. Critical evidence in this debate typically consists of comparisons between the perception of speech and non-speech by humans as well as the perception of speech by humans and non-human animals.

Non-speech signals mimic the speech signal in certain crucial respects. For example, while non-speech is not perceived as speech, it can be created to resemble speech in terms of rapid changes in frequency and amplitude. Experiments on non-speech test the claim that speech-specific mechanisms are required for the perception of speech. Clearly, this claim is weakened by any finding that non-speech is perceived in a way similar to speech, supporting a general auditory mechanism

model for the perception of speech. Likewise, any finding that non-human animals perceive speech like humans would also strengthen support for the notion that speech perception involves the use of general auditory mechanisms across species. We will briefly consider research in these two areas.

13.3.2.1 NON-SPEECH PERCEPTION

Non-speech versions have been created for many linguistic contrasts, including consonant voicing, place, and manner of articulation. For example, in creating a non-speech analog of a plosive voicing continuum, consider that VOT expresses the temporal relationship between two events, the release of the burst and the onset of voicing. A non-speech version of such a VOT continuum can be created by systematically varying the interval between two acoustic events. Figure 13.7 illustrates a few members of a tone onset time (TOT) continuum in which the relative onset of two pure tones was manipulated. This continuum mimics the crucial properties of a VOT continuum (the temporal relationship between two events) without being perceived as speech.

Research has shown that listeners perceive these types of non-speech continua in a way that is very similar to the perception of their speech counterparts. For example, Pisoni (1977) found similar categorical perception results for listeners categorizing VOT and TOT continua. Similar findings have also been reported for non-speech continua modeled after distinctions in terms of manner of articulation (e.g. /b-w/ and /t-l/) and place of articulation (e.g. /d-g/), suggesting a general mechanisms account of speech processing.

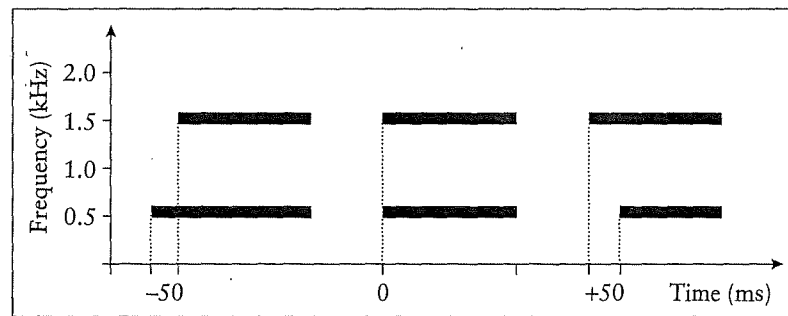


Figure 13.7 Stimuli from Pisoni (1977). Schematic representation of three of the stimuli used in a tone onset time analog of a VOT continuum. Each stimulus consists of a 500 Hz and a 1500 Hz tone. The relative onset of the two tones is systematically varied. The left panel shows the lower tone leading the higher tone by 50 ms, simulating a -50 ms VOT. In the middle panel, both tones start simultaneously, simulating a 0 ms VOT. In the right panel, the lower tone lags the higher tone by 50 ms, simulating a +50 ms VOT.

13.3.2.2 ANIMAL PERCEPTION

Early studies on categorical perception in non-human animals were conducted by Kuhl and Miller (1975). These researchers compared the perception of VOT continua by humans and chinchillas. The chinchillas (furry mammals from Latin America whose auditory system is quite similar to that of humans) were first trained to identify the endpoints of the continuum using shock avoidance. That is, in response to one of the endpoint stimuli, the chinchillas were taught to avoid mild electric shock by moving from one half of their cage to the other half. For the other endpoint, the chinchillas were trained to stay put and inhibit this crossing response. Correct inhibition was rewarded with drinking water. Once they had been taught to distinguish these stimuli, the remaining members of the continuum were included for identification. Results showed that the chinchillas' voicing boundary was very similar to the humans'. Kuhl and Miller compared perception of the voicing distinction at three places of articulation: bilabial, alveolar, and velar. They found a very close correspondence in that as the human VOT boundary shifted to increasingly longer VOTs at each place of articulation, so did the animal boundary.

Studies on animal perception have included additional linguistic contrasts (e.g. place and manner of articulation) as well as a variety of species (e.g. rhesus monkeys, macaques, and Japanese quail). In general, results from these studies indicate that there is a close correspondence between the ways in which humans and animals perceive linguistic distinctions.

What do results from experiments on the perception of non-speech and the perception of speech by animals tell us? Caution is required when interpreting these results. The demonstration that non-speech is perceived in a way similar to speech seems to indicate that an account of basic speech perception processes does not need to invoke the notion of speech-specific mechanisms. General auditory processes can account for the perception of both speech and other auditory signals. Likewise, the finding that animals perceive speech like humans indicates that basic speech perception phenomena can be understood in terms of the use of general auditory mechanisms. For example, it has been hypothesized that a voicing boundary at a VOT of 20 ms or more directly reflects the temporal resolution of the auditory system. Hirsh (1959) showed that listeners can only judge the temporal order of two auditory events when they are separated by at least 20 ms. If the onsets of the two events occur within 20 ms of each other, listeners are unable to determine which event occurred first. Translated to VOT (or TOT), one event is the release of the burst; the other is the onset of voicing. The implication of Hirsh's finding is that if release and onset of voicing occur within 20 ms of each other, the listener will treat them as occurring simultaneously, which results in the perception of a voiced plosive for a language like English. If, however, the onset of voicing occurs 20 ms or more after the release, the listener can distinguish two separate events and tell that voicing lagged release, which results in a voiceless percept. Thus, the voicing boundary could stem from general properties

and limitations of the auditory system. Indeed, recent research with non-speech indicates that basic auditory sensitivities affect the extent to which categories can be learned: participants more quickly learned to distinguish two distributions when the boundary between them fell at a TOT of 20 ms as compared to two distributions whose boundary occurred at a TOT of 40 ms. The latter distinction presumably was more difficult to learn because a 40 ms TOT does not correspond to any known peak in discrimination and because one of the distributions was centered around a 20 ms TOT, which is a known area of increased sensitivity (Holt et al. 2004). This finding suggests that learnability may shape language sound inventories.

Finally, we do not claim that animals will be able to perform just about every speech or non-speech perception task that humans do. One possible area where the close correspondence between human and animal performance may break down is that of higher-level equivalence classification. For example, while infants have been shown to detect a correspondence between the vowel sounds /a/ or /i/ and the sight of a person producing them, a similar pattern in animals has not been reported.

13.4 The Role of Linguistic Experience in Speech Perception

So far, we have assumed that speaker and hearer speak the same native language. However, in many cultures, bilingualism or trilingualism is the norm; speakers may be equally proficient in both or all three languages, or have one dominant language. In addition, many people leave their native countries for ecological, political, educational, economic, or even touristic reasons. As a result, conversations in which some or all participants share a second language are always common in many areas of the world. Research on non-native production and perception of speech constitutes a relatively new area, known as the **phonetics of second language acquisition**. The study of non-native speech perception proper is often referred to as the study of **cross-language speech perception**. We will briefly review some of the issues in this area.

When an adult learns a second language (L2), the native language (L1) has already been firmly established over a number of years. A basic issue therefore concerns the extent to which this L1 system influences the perception of the L2. Early work on cross-language perception was dominated by the categorical perception paradigm. Research explored whether listeners with different native languages show any language-dependent differences in their placement of the voicing boundary along a VOT continuum. Specifically, a wide-ranging /ba/-/pa/ VOT continuum from -150 ms to +150 ms was presented to native speakers of English, Spanish, and Thai. As discussed in Section 6.3, word-initially, English contrasts plosives

with a negative or short positive VOT to plosives with a long positive VOT, Spanish contrasts plosives with a negative VOT to plosives with a short positive VOT, and Thai makes a three-way distinction between plosives with a negative VOT, a short positive VOT, and a long positive VOT. Identification results showed that, when presented with the identical continuum, English listeners placed the bilabial voicing boundary at +20 ms, Spanish listeners at -5 ms, and Thai listeners showed two boundaries, one at -20 ms, the other at +40 ms. Results from additional discrimination experiments also showed that listeners only distinguished well between stimuli from different categories. English and Spanish listeners each showed two distinct categories and Thai listeners showed three distinct categories. In sum, the way in which listeners divide a continuum is determined by language-specific experience (Abramson and Lisker 1970; Williams 1977). Subsequent studies corroborated these findings for a variety of languages and consonantal distinctions.

The results raise the question of whether these perceptual patterns change as listeners gain more experience in the L2.

The ability to override or adapt L1 settings when processing L2 is thought to reflect the brain's **plasticity**, that is, its potential to reorganize its structure and function in response to experience. The notion of plasticity is closely tied to the **critical period hypothesis**, according to which there exists a particular window of opportunity during which native proficiency in a language can be obtained (Lenneberg 1967). There is as yet no consensus about the existence or duration of this critical period, but it is clear that there is a high degree of plasticity in the perceptual system, not only in infants and children, but in adults as well. Under optimal circumstances for L2 learning – that is, when L2 exposure begins at a very early age – near-native or native proficiency can be obtained. Research along these lines forms part of the study of **bilingualism**. However, research has shown that even L2 learners who have managed to distinguish difficult non-native contrasts may not use the same acoustic cues as natives (Underbakke et al. 1988).

For older children, modification of L1 patterns is possible with sufficient exposure. Research has shown that the more time Spanish-speaking children (ages 8–10 and 14–16) spent in the US, the more English-like their voicing boundary became. However, even after three years in the US, the Spanish children's voicing boundary still occurred at a substantially shorter VOT than that of monolingual American children (Williams 1979). For adults as well, perception can be substantially improved by means of a brief training regimen that focuses on a specific difficult L2 phoneme contrast. Ten to twenty hours of training in which participants are exposed to target speech sounds produced by a variety of speakers and in different phonetic contexts yield substantial improvement in participants' ability to perceive and even to produce difficult contrasts (Logan et al. 1991; Wang et al. 1999).

While children and adults show some malleability in their phonetic categories, infants as young as 2–4 months are sensitive to a great variety of subtle acoustic differences, regardless of whether these differences serve to differentiate phonemes

in their ambient language. That is, infants categorically discriminate among the phonemes of their native language as well as many other non-native phonemes. This sensitivity, however, seems to weaken at the end of the first year of life. Thus, the native language acts as a filter in the first 12 months that emphasizes native sound contrasts while de-emphasizing non-native ones. For example, infants from both a Hindi- and an English-speaking environment discriminate a dental /t/ from a retroflex /ɖ/, two sounds that contrast phonemically in Hindi but not in English. However, while adult speakers of Hindi continue to distinguish these two sounds, adult speakers of English do not. It turns out that at age 10–12 months, English-speaking infants no longer discriminate the two sounds (Werker and Tees 1984; see Cheour et al. 1998 for a similar conclusion based on EEG).

Results from the perception of vowels show an influence of the native language at an even earlier age. In her research on the “perceptual magnet effect”, Kuhl (Grieser and Kuhl 1989; Kuhl 1991) showed that both adults and infants find it more difficult to distinguish a prototypical instance of the vowel /i/ from variants that surround it in acoustic space as compared to distinguishing a non-prototypical instance of the vowel /i/ from its variants, even though the auditory distance between (non)prototype and variants is the same for both conditions. In Kuhl’s terms, the prototype acts as a magnet, attracting other members of the category and thereby shrinking the space. Subsequently, this magnet effect was shown to be determined by the ambient language: American 6-month-olds showed a magnet effect for prototypical English /i/ but not for prototypical Swedish /y/. In contrast, Swedish 6-month-olds showed a magnet effect for prototypical Swedish /y/ but not for English /i/ (Kuhl et al. 1992). Thus, for vowel perception, the effect of language experience can be detected as early as 6 months.

Returning to adults, it is not the case that monolingual adults cannot distinguish among any non-native phonemes. While some non-native contrasts are discriminated poorly, others are discriminated at near native-like levels. It has been shown that the relation between the sound inventories of the L1 and L2 is a major predictor of the way in which an L2 phoneme will be perceived. There are currently two major models of L2 phonetic category acquisition, the perceptual assimilation model (PAM; Best 1994) and the speech learning model (SLM; Flege 1995). As emphasized by Best and Tyler (2007), PAM is a model of non-native speech perception by naïve listeners unfamiliar with the target language. Non-native contrasts are perceived in terms of their gestural similarity to the phonological categories in the native language. Consideration of the way in which the members of a contrastive pair of phones are perceptually assimilated allows PAM to make specific predictions about discrimination performance. Very good discrimination should obtain when the two non-native phones are perceived as exemplars of two different native phonemes (“two category” assimilation). Poor discrimination occurs when the two non-native phones are perceived as equally good (or poor) instances of the same native phoneme (“single category” assimilation). Intermediate discrimination is predicted when two non-native phones are perceived as instances of the same native phoneme but with different

category goodness ratings ("category goodness" difference). In addition to these different degrees of assimilation, it is possible that one or both members of the pair are not perceived as instances of any native phoneme (are "uncategorized"). PAM predicts very good discrimination for categorized-uncategorized contrasts while discrimination of two uncategorized phones depends on the extent to which each phone maps on to a distinct native phoneme.

A classic example is the acquisition of the distinction between /ɪ/ and /I/ by Japanese learners of English. While Japanese has a sound resembling /ɪ/ (which is often realized as a retroflex tap /ɾ/) it does not have /I/ and Japanese learners have difficulty distinguishing the two (Strange 1992). According to PAM, this difficulty arises from the fact that both /ɪ/ and /I/ are treated as poor instances of the native Japanese /ɪ/ category, an instance of single-category assimilation. In contrast, native English speakers accurately distinguish the Zulu distinction between the voiceless lateral fricative /ɬ/ and its voiced counterpart /ɮ/. While neither fricative occurs in English, this is an example of two-category assimilation whereby English speakers map Zulu /ɬ/ onto English /s/ and Zulu /ɮ/ onto English /l/ (Best et al. 2001).

SLM is specifically concerned with ultimate attainment of L2 pronunciation and posits that errors in L2 production mostly have a perceptual basis. Like PAM, the relation between sounds in the native language and L2 plays an important role in the perception of non-native sounds. Unlike PAM, SLM does not specify whether this similarity is acoustic or articulatory in nature. SLM makes a distinction between "new" and "similar" phones. New L2 phones have no counterpart in the native language and therefore differ from all L1 phones. For example, the high front rounded French vowel [y] is a new phone for English learners of French. Similar L2 phones resemble but are not identical to L1 counterparts. Consider French /t/: although French /t/ is not identical to English /t/ since it is dental and unaspirated, it is clearly similar to English /t/. SLM predicts that the process of equivalence classification prevents L2 learners from creating a new category for similar, but not new, sounds. Thus, L2 learners are predicted not to achieve native-like pronunciation for similar phones. However, given sufficient exposure, L2 learners should be able to produce new phones authentically.

Research by Flege and colleagues has shown that there is indeed a difference in the acquisition of new and similar phones (e.g. Flege 1987). Advanced English learners of French mastered French /y/, a new phone, but not French /u/, a similar phone. Interestingly, not only do L1 categories affect L2 sounds: acquisition of L2 categories affects the production of L1 sounds as well. For example, native French speakers who had been living in the US for more than 10 years produced both English and French /t/ with VOT values in between the native norms for those two languages (Flege 1987).

For both PAM and SLM, the notion of phonetic similarity is crucial in predicting how listeners will treat L2 sounds. However, a clear empirical test for the assessment of phonetic similarity that is independent of L2 perceptual difficulties remains to be developed (see Strange 2007).

In sum, monolingual adult listeners have difficulty perceiving many non-native contrasts. However, perception is not uniformly poor. The perceptual difficulty that the adult listener encounters cannot be predicted solely on the basis of a comparison between the L1 and L2 phoneme inventories. Instead, research indicates that several other factors need to be taken into account, including the specific acoustic cues to mark a phonemic distinction, the allophonic distribution of phonetic segments in L1 and L2, the age of the speaker, as well as the amount of exposure to the L2, the quality of the L2 input, and the extent of usage of both the L1 and L2. Nevertheless, the perceptual system exhibits considerable plasticity such that phonetic categories can be modified even in adults.

13.5 Summary

The process by which listeners extract information from the speech signal, allowing them to comprehend the speaker's message, is complex and as yet not fully understood. While there is a reasonably good understanding of the acoustic cues to individual segments, there is debate as to whether these cues are variable or stable. Also, consensus on the basic units of speech perception is still lacking. An additional issue concerns the question of whether speech perception is based on general auditory mechanisms or on mechanisms that evolved especially for the processing of speech. Finally, while the perceptual system displays considerable plasticity, our native language affects the way in which we perceive speech.

Exercises

- 1 What are the arguments for and against extrinsic normalization? How does extrinsic normalization differ from point and range normalization?
- 2 What are the perceptual cues to place of articulation for fricatives? Where in a syllable are these cues located?
- 3 What are the perceptual cues to place and manner of articulation for plosives? Which cues are context-dependent? Give examples.
- 4 What is the motor theory of speech perception? In your description, include evidence supporting and opposing the theory.
- 5 What two types of experiments must be conducted in order to test categorical perception? Illustrate the use of these experiments with a "ba" to "pa" continuum. How does categorical perception differ from continuous perception?
- 6 What evidence is there that categorical perception is an innate feature of human language? How was that evidence collected? What evidence is there to the contrary?

- 7 What are the perceptual assimilation model and the speech learning model? How are they different? Would either or both of these models predict that American learners of Korean would have difficulty with the contrast between the rounded back vowel [u] and the unrounded back vowel [ʊ]? Explain your answer.

Note

- 1 Two corresponding pieces of reel-to-reel audio tape with the relevant signal portions were swapped with each other and joined together. Nowadays, splicing is done digitally, using a "cut and paste" action on the computer.